

IOT ENHANCED ROBOTIC WASTE COLLECTOR WITH BLYNK APP CONTROL AND METAL PROXIMITY SENSING

A Project Report

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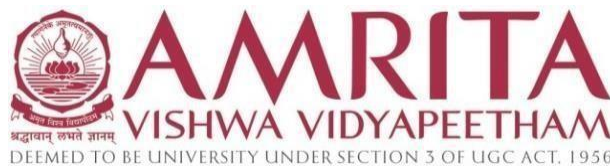
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under the supervision of

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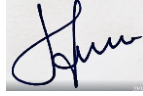


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Certificate

This is to certify that this project titled **“IoT Enhanced Robotic Waste Collector with Blynk App Control and Metal Proximity Sensing”** submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer and Communication Engineering**, by **Aksshaya R, Dorai Karthikeyan G M, Rangashree Dhanvanth and Swathi P**, is a bonafide record of work carried out by them, under my supervision and that it has not been submitted, to the best of my knowledge, in part or in full, for the award of any other degree or diploma.



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We do hereby declare that this project titled **“IoT Enhanced Robotic Waste Collector with Blynk App Control and Metal Proximity Sensing”**, submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer & Communication Engineering**, is a true record of work carried out by us, under the supervision of **Mr. Ganesan M** and that all information contained herein, which do not arise directly from our work, have been properly acknowledged and cited, using acceptable international standards. Further, we declare that the contents of this thesis have not been submitted, in part or in full, for the award of any other degree or diploma.

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Dedicated to
Our parents, teachers, Friends, and well-wishers,
Who have encouraged and supported us throughout our life.

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Abstract

When it comes to trash management, efficient metal waste collection, and recycling are essential to reducing environmental risks. Metals make up a large amount of waste, and the ecosystem is seriously endangered by unnecessary disposal. It takes a lot of time to post-segregate waste by conventional methods, especially when dealing with metal waste. To reduce the negative impacts of metal trash in landfills, creative ways to expedite and enhance the sorting process are needed.

Our approach is based on a comprehensive view of waste handling, utilizing a self-contained robotic system intended for effective material separation. The fundamental operation of the system, enabled by the ESP32 microprocessor, depends on the efficient recognition and classification of materials according to their metallic characteristics. By improving the system's ability to distinguish between metallic and non-metallic materials, the metal proximity sensor guarantees careful material sorting. Additionally, the incorporation of the Blynk Android application, which offers an easy-to-use interface for controlling and seeing the robot's activities in real-time, facilitates user involvement. This collaborative hardware-software ecosystem enables users to direct the robot in depositing materials into designated bins categorized by material type, streamlining waste disposal and ensuring materials are appropriately sorted for subsequent recycling efforts.

In conclusion, this ESP32 microprocessor-powered robotic waste management system provides a creative way to speed up and improve the sorting of metal waste while lowering environmental hazards. The thorough sorting of materials is made easier by the effective identification and classification of materials utilizing metal proximity sensors, which are essential in differentiating between metallic and non-metallic substances. By integrating the Blynk Android application, user engagement is streamlined, and real-time control and monitoring of the robotic system is made possible. This approach contributes to environmentally friendly waste management techniques while streamlining garbage disposal and guaranteeing efficient material segregation, opening the door for more long-term recycling initiatives.

Table of Contents

List of Abbreviations	i
List of Symbols	ii
List of Tables	iii
List of Figures	iv
1. Introduction	1
1.1 Background	1
1.1.1 Current Challenges in Waste Management	1
1.1.2 Environmental Impact of Uncollected Waste	1
1.2 Role of IoT in Enhancing Waste Management	2
1.2.1 Automation in Waste Segregation	2
1.2.2 Integration of IoT and Blynk for Smart Control	2
1.2.3 Existing vs Robotic Waste Management Technologies	2
1.2.4 Contribution to Sustainable Practices	3
2. Literature Survey	5
3. Methodology	7
3.1 System Architecture	7
3.1.1 Overview of Robotic System Components	8
3.1.2 Hardware components Requirements	8
3.1.3 Software requirements	11
3.2 Working Modes	12
3.2.1 Automatic Mode	12
3.2.2 Manual Mode	14
3.3 Steps Involved	15
3.3.1 Trash Collection and Segregation	15
3.3.2 Bin Level Monitoring Using Blynk App	17
4. Result and Discussion	18
4.1 Simulation Results	18
4.2 Sensor Calibration	19
4.3 Robot Chassis Setup	20
4.4 Blynk App Interface	21

4.5 Integration of Manipulator with Robot Chassis	24
4.6 Trails Conducted for Waste Collection and Segregation	25
4.7 Challenges Encountered	27
5. Conclusion and Future Scope	29
Bibliography	30
Publications	33

List of Abbreviations

BO	Battery Operated
IDE	Integrated Development Environment
IoT	Internet of Things
Rx	Receiver
Tx	Transmitter
Wi-Fi	Wireless Fidelity

List of Symbols

cm	Centimeter
$^{\circ}$	Degree
d	Distance
s	speed
t	Time
V	Voltage

List of Tables

Table No.	Table Caption	Page No.
Table 3-1	Materials Detected by Inductive Sensor	16
Table 4-1	Calibrated Sensor Values	19
Table 4-2	Trails for manual waste collection	25
Table 4-3	Trails of dropping waste into bin	25
Table 4-4	Trails of segregation in manual mode	25
Table 4-5	Trails of automatic waste collection	26
Table 4-6	Trails of dropping waste into specific bin	26
Table 4-7	Trails of segregation in automatic mode	26

List of Figures

Figure No.	Figure Caption	Page No.
Fig. 3.1	The Block Diagram	7
Fig. 3.2	Structure of Robotic Waste Collector	8
Fig. 3.3	Flowchart for Autonomous Movement of Robot	14
Fig. 3.4	Flowchart for Trash Identification	16
Fig. 4.1	Simulation of BO and Servo Motors	18
Fig. 4.2	Virtual Terminal for Wheel Rotation	19
Fig. 4.3	Virtual Terminal for Pick, Carry, Rotate	19
Fig. 4.4	Robot Chassis setup – Front view	21
Fig. 4.5	Robot Chassis setup – Side view	21
Fig. 4.6	Blynk Application Interface	23
Fig. 4.7	Resting Position of Robotic Waste collector	24
Fig. 4.8	Collecting of waste using Manipulator	24

1. Introduction

Maintaining public health and environmental sustainability both depend on effective waste management. The amount of garbage produced has increased due to ongoing industrialization and population growth, which poses a serious problem for businesses and communities around the world. This section provides an overview of waste management in a broader framework, highlighting the need for practical solutions to today's problems and reducing the harm that uncollected waste causes to the environment.

1.1 Background

Waste is an unavoidable consequence of human activity in the commercial, industrial, and domestic domains. The amount and variety of trash created has increased to an unprecedented degree as a result of the acceleration of industrial expansion and urbanization. Modern waste streams are so large and complicated that they are beyond the capacity of traditional waste management methods [1], which usually depend on dumping and burning.

1.1.1 Current Challenges in Waste Management

The primary challenge in today's garbage or metal trash collection is the traditional method of gathering waste at one spot and then delivering it to a specific unit for trash segregation or separation. The two stages of this conventional procedure are gathering all trash kinds in one location and then sorting the garbage according to whether it is metallic or non-metallic. This methodical approach has some shortcomings and inefficiencies. First off, there is an environmental impact and an increase in operational costs due to the additional resources required for the transportation of mixed trash to a segregation unit. Furthermore, there is a delay before the actual sorting process starts due to the time it takes to transport garbage to a different facility for segregation.

1.1.2 Environmental Impact of Uncollected Waste

Uncollected metal waste poses a serious threat to the environment that has far-reaching effects. By the year 2025, the world's annual production of metal waste is expected to reach 2 billion metric tons, which will contaminate the land and water. Metals like lead and mercury leak into the environment due to improper disposal and a lack of an effective waste management infrastructure. This puts human health in danger and puts ecosystems at risk of long-term harm and diseases. Furthermore, according to estimates from the World Health Organization, environmental hazards—including those related to unregulated metal waste—cause more than 12 million deaths yearly. Immediate and creative action is needed

to improve garbage collection, segregation, and recycling procedures worldwide to address this problem.

1.2 Role of IoT in Enhancing Waste Management

Waste management is going through a new era as a result of the integration of the Internet of Things (IoT) [1].

1.2.1 Automation in Waste Segregation

Waste segregation procedures are greatly improved by the automation that comes with integrating IoT into waste management systems [1]. Waste collection and sorting have grown more efficient and intelligent with the use of smart sensors and networked devices. Smart Sensors for Material Identification: Metal proximity sensors [2] and other IoT-enabled smart sensors help automate the trash sorting process. These sensors can differentiate between metallic and non-metallic components [3][4] when determining the material of waste products. The incorporation of these sensors into the robotic arms of our trash collector enables real-time junk detection.

Efficient On-Site Segregation: The Internet of Things-enabled automated trash segregation method reduces the need for post-collection segregation [5]. The waste collector can reduce the time and resources needed for sorting by performing on-site segregation [5] using smart sensors. This ultimately results in a waste management [1] workflow that is quicker and more effective to transport garbage to a different facility for segregation.

1.2.2 Integration of IoT and Blynk for Smart Control

The interplay between the Blynk platform [6] [7] and the Internet of Things improves the intelligent control capabilities of waste management systems [1]. ESP32 Microcontroller for Connectivity: With integrated Wi-Fi and Bluetooth modules, the ESP32 microcontroller [8] is essential for the garbage collector to establish connectivity. With the help of this microcontroller, the garbage collector may connect to the Internet of Things ecosystem and exchange and regulate data in real-time.

Blynk Application for User Interaction: The flexible IoT platform Blynk is used to create an intuitive waste collection interface. Users may remotely operate and keep an eye on the garbage collector in both manual and automatic modes using the Blynk application [6][7]. User involvement is improved by the user-friendly interface, which makes it easy for people to control the robot, keep an eye on bin levels, and get alerts [9].

1.2.3 Existing vs Robotic Waste Management Technologies

This project aims to establish the importance of proper waste disposal for the well-being of the public and the conservation of the earth's environment. The common methods of dealing with trash such as collection, landfilling, incineration, and recycling pose problems and challenges. For instance, conventional waste collection involves sorting to separate the waste, which is tiresome, time-consuming, and makes the waste contaminated thereby reducing the recycling rates. Improper disposal leads to the pollution of the soil and groundwater. While centers engaged in the recycling of certain kinds of waste help in the processing of wastes, the efficiency level largely depends on the quality of segregation of wastes, and this is exacerbated by issues arising from contamination.

In contrast to traditional methods, the Robotic waste collector's new on-site waste segregation system introduces the concept of IoT with robotic integration that has not been used in the previous methods. At the collection level, the ongoing segregation is made through metal proximity sensors that enable the system to segregate waste as it is collected. This approach helps to save time and energy when sorting waste manually and provides higher-quality recyclable products. Furthermore, on-site segregation aids in realizing better recycling percentages that will also help in minimizing the impact on the environment as well as save resources due to proper processing of metals and other materials. Considering metals as materials that can be infinitely recycled and used for countless purposes in other industries. Furthermore, keeping metals separate from general waste prevents the release of harmful substances during landfill decomposition or burning, reducing pollution to the environment and health concerns.

Moreover, our robotic system is also armed with an inclusive interface of the Blynk app which provides instant monitoring and control features. The users can know the state of waste bins, observe the separation process, and receive notifications about overfilled bins. Therefore, deploying IoT in a robotic waste collector with Blynk app control system and metal proximity sensor proves to be more efficient, cheaper, and more environmentally friendly compared to conventional means.

1.2.4 Contribution to Sustainable Practices

To sustain management of wastes from human activities, the following below practices can be adopted; This makes the plan of developing an IoT-Integrated Robotic Waste Collector in line with the notion of sustainability in addressing issues concerning the practice of sustainable development because it focuses on improving the mechanisms of

waste collection. The advantages which could be accrued for the implementation of this robotic system include returns which increase sustainability.

Leads to reduction of environmental impact. The normal practices of garbage disposal and segregation would involve so much human efforts and the use of many vehicles hence causing much harm to the environment through the release of carbon and much energy usage. The IoT-enhanced robotic waste collection mitigates these effects by the IoT-enhanced robotic waste collection mitigates these effects by Minimizing Transportation Needs, Waste segregation at the point of collection reduces the requirement to transport mixed waste to central facilities, resulting in lower fuel consumption and emissions.

Motor power and the sensors used by the robot are low power and much less than other standard garbage management vehicles making it energy efficient. The use of a metal proximity sensor enhances recycling efficiency of the garbage collector and recycles metals in the shortest time possible. This promotes sustainable practices by This promotes sustainable practices by:

Automated segregation helps to avoid situations when certain types of wastes are not collected at all, or collected and sorted in a wrong manner, thus increasing the efficiency of recycling. Source separation is essential since contamination during the recycling process lowers the quality of the recycled products. This initiative benefits the ideas of the circular economy by enhancing the separation of waste seeking to incorporate them into various processes as much as possible. The benefits include resource recovery.

Separation of recyclable material from the waste limits the amounts of waste that is publicly dumped, hence increasing sustainability in the long-term usage. Through IoT, the system's components can prompt efficient waste management practices by collecting and monitoring relevant real-time data. Monitoring necessary stock levels in the bins in real-time mode contributes to the optimization of collection routes and efficient storage management, which will eliminate unnecessary traveling and emissions. Supervising the status of operation of the robot can be used to forecast the need for maintenance, and this way, the functioning of the system can be maintained optimally with the least amount of time being down, thus saving on resource utilization helping in predictive maintenance.

2. Literature Survey

IoT is transforming remote monitoring and control, addressing key issues like solid waste management. This advanced system improves efficiency, reduce waste, and enhance sustainability. By changing how we manage and monitor waste, IoT significantly benefits our communities and creates a more sustainable future [1]. With the rapid population growth, automation is essential for making life easier and more comfortable. Advances in automation and embedded systems are driving this shift, streamlining daily tasks and improving overall efficiency [2]. Automation not just saves time but also saves human effort and labor. Especially automation helps in industries like healthcare and waste management systems. Through this automation, it reduces the risk of humans being directly affected by various diseases which are the root causes from picking up the waste manually. [3] states that segregating waste at the source is crucial for effective recycling. Metallic waste can be reused or recycled more efficiently when sorted early, ensuring higher material quality and greater value recovery. While industrial segregators exist, source segregation reduces occupational hazards for waste workers and streamlines the recycling process. Directly sending segregated waste to recycling plants bypasses the need for additional segregation, enhancing overall efficiency and sustainability in waste management. This approach not only maximizes the value extracted from waste but also improves safety and operational processes in the recycling industry. [4] incorporates a key aspect of sustainable development is designing and implementing systems for efficient waste collection and removal. Bin level monitoring plays a crucial role in this process by tracking waste levels in real time, ensuring timely collection, reducing overflow, and optimizing waste management operations for a cleaner, more sustainable environment. This type of bin level monitoring can be done by using a mobile application called Blynk software platform. [5] proposes Blynk is a software application for creating hardware control and monitoring interfaces. In our project, Blynk assists to check bin levels and manage the mobility of robots with the help of signals coming from mobile phones.

For the purpose of detecting the metallic scrap, which is picked up by the robot, we install a proximity sensor that is specifically a metal proximity sensor also known as an inductive proximity sensor. This sensor is used to detect metallic objects in the surroundings and does not need to encounter it. It utilizes electromagnetic induction to measure the modification of the field generated by the metallic objects; it immediately informs the system of the presence of metal. Thus, introducing the metal proximity sensor into the project will help ensure that the incorporated robot sorts the materials into metallic

and non-metallic waste. Due to this capability, waste management becomes efficient in terms of workflow and the number of mistakes committed is minimized. Also, because the current type of sensor is non-contact, it ensures a safe and long operation because direct contact with metallic objects can easily cause shorting or breaking [6]. The proximity sensor used for waste segregation is integrated in the ESP32 microcontroller. Due to its features ESP32 microcontroller is integrated in IoT projects such as it has a dual-core CPU which can be operated at a different frequency of 80, 160 or 240 MHz, it offers powerful processing capabilities. This allows the ESP32 to handle complex activities including running camera interfaces and speech recognition alongside relaying information from the internet at the same time. Another advantage of the ESP32 is that it incorporates Wi-Fi and Bluetooth interfaces, thus negating the usual necessity for extra radio modules as seen in other microcontrollers including those from Arduino. This facilitates project development and cuts the cost of components hence making ESP32 suitable for IoT projects. It is powerful and flexible and can therefore be used in any kind of application ranging from home automation to industrial monitoring applications [7][8]. The Blynk app, other than monitoring the bin levels, also notifies the user once the bin is full, which helps in timely disposal of waste and doesn't require any type of human interaction. Humans from anywhere in the country be aware of the robot's bin level [9]. The HC-SR04 ultrasonic sensor is used to determine distance to an object. It offers non-contact range detection from about 2 cm to 400 cm, depending on the type of ultrasonic sensor used. The ultrasonic sensor emits the short and high frequency signal [10]. If they detect any object, then they reflect echo signal which is taken as input to the sensor through Echo pin. In this project ultrasonic sensor is used for obstacle detection and trash detection. The ultrasonic sensor placed closer to the chassis detects the trash while the one placed closer to the arm of the manipulator detects the obstacle. Servo motor is also integrated with the ESP32 microcontroller and is used for various purposes in this project like bin rotation, picking and placing of the trash by the manipulator. Bin rotation is used when classifying the metallic characteristics of the trash [11]. Namratha in [12] says that "Recycling is the best way to create a sustainable environment and also it needs the segregation of waste materials which is a tedious time-consuming task"

3. Methodology

The primary goal of this robot is to take trash collected at the workplace and parks to determine whether they are metal or not, and then place them in the respective bins so that the collected trash can eventually be sent for recycling.

3.1 System Architecture

Fig. 3.1 depicts the diagram of our robotic waste collector which is made up of motors, sensors, a micro-controller, and a metal proximity sensor.

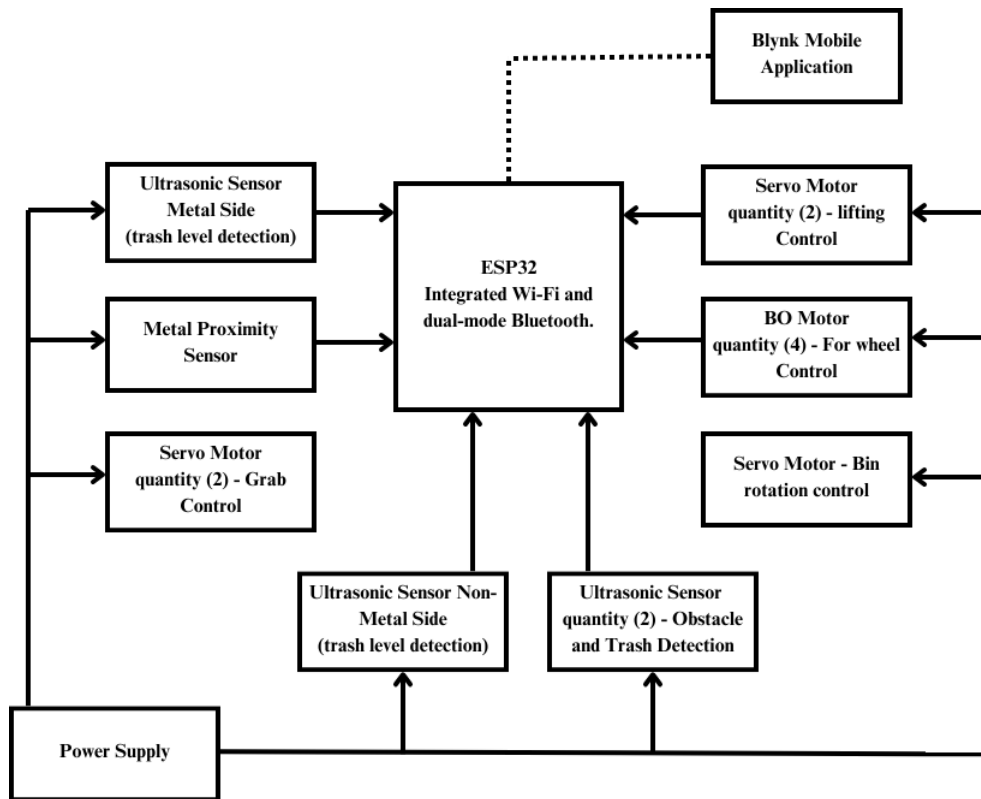


Fig. 3.1 The Block Diagram

All the components in our robotic system are managed and controlled by the ESP32 microcontroller [8] which has an Integrated Wi-Fi and dual-mode Bluetooth. The power supply for the entire robot is given by a 12v Lithium-Ion battery. The robotic model has a bin attached to it which is divided into two sections, one for metal trash and another for non-metal trash. The movement of the robot is achieved by 4 BO motors attached to the wheels. The robot has 2 ultrasonic sensors in front of the chassis for obstacle detection [10] [11]. The robot features two robotic manipulators. Two servo motors control the arm's sideways movement as it grabs the rubbish. The arm's grabbing mechanism includes a metal inductive sensor that determines if the junk seized is metal or non-metal. The bin rotates with the help of the sensor. The servo motor [12] can rotate 180° based on the detection by

the Metal proximity sensor. Two more servo motors regulate the arm's up and down movement to place the trash in the proper portion of the bin. Each part of the bin contains ultrasonic sensors that continuously check the bin's level. The levels are displayed in the Blynk app for users. The overall structure of the robot is depicted in Fig. 3.2.



Fig. 3.2 Structure of Robotic Waste Collector [5]

3.1.1 Overview of Robotic System Components

The components used for the robotic waste collector can be categorized into hardware and software requirement.

3.1.2 Hardware components Requirements

(a) ESP32-WROOM-32

The ESP32 controls most of the operations on the robotic waste collector. Picking this flexible microcontroller was based on the fact that it is richly featured, high-performance, and possesses suitable interfacing options, which are crucial for this project. ESP32 is a very powerful microcontroller which has two processors, multiple input/output ports, as well as on board Wi-Fi and Bluetooth interfaces. These characteristics make it suitable to be used in IoT applications as well as large projects where stable wireless performance and computing power are needed.

Another of the fundamental reasons for selecting the ESP32 microcontroller model was the large number of input/output pins. This function is critical for the robotic garbage collector, which involves numerous sensor connections and motor controls. The ESP32 can connect to multiple sensors such as ultrasonic sensors for the detection of objects or obstacles and to measure the level of bin, metal proximity sensor to identify metals. Since the ESP32 has several GPIO pins, easy control of multiple BO motors and servo motors is possible to produce the motions and functioning of the required segments of the robot.

ESP32 has a built-in Wi-Fi module. The built-in Wi-Fi module allows creating a wireless connection with the device. This feature enables several important components of the robotic garbage to operate via Blynk App; therefore, the user can operate the robot autonomously as well as monitor and control it remotely using manual control. This link provides the facility for real time changes and manipulations of the actions of the robot.

The ESP32 receives sensor data from environments or device operational status and transmits to cloud or another server. Due to response real-time data transmission, it is used to assess the performance of the system as well as data on best garbage collection optimization. The ESP32 has dual core processor, and it offers a good amount of computational power, which is necessary for performing the various operations in the robotic garbage collector. ESP32 also helps in parallel processing. The breakdown of tasks that the microcontroller can perform at the same time include reading data from the sensors, controlling motors, and incorporating Wi-Fi connectivity in one single chip without compromising on the system's speed. The processing power allows efficient task management, quick responses, which are important in real-time operations and immediate actions, especially in the conditions of dynamic garbage collection.

(b) Bo Motor

The BO Motor is used to operate the wheels on the robotic waste collector. It provides the required torque that helps in the movement of the robot across surfaces even with load. It also provides precise control of the speed of the robot and enables acceleration and deceleration. This provides the mechanical power needed for the movement of the robot through the conversion of its electrical energy to mechanical energy. BO motors propel the robot, allowing it to move forward, backward, turn, and navigate in response to user commands or autonomously avoid obstacles.

(c) L293D Motor Driver

The L293D Motor Driver regulates voltage and current to control the speed and direction of BO motors in the robotic system. It interfaces the microcontroller with high-power motors. By managing current flow, it prevents motor damage. Its features include a dual H-bridge configuration with the ability to control two DC motors simultaneously with bidirectional control.

(d) Servo Motor

The Servo Motor is designed for precise angular control and is used for high Torque Movement Control. It can be controlled by pulse width modulation (PWM) signals from the

ESP32 microcontroller. In the robotic waste collector, SG90 and MG996R servo motors have been used for bin rotation and picking and carrying waste respectively. The Servo motor's rotation range is 180°. It is used for collecting waste from the ground and placing the waste into the bin, and arm elevation [12]. The SG90 servo motor is used for the rotation of the bin based on metal or non-metal waste collected.

(e) HC-SR04 Ultrasonic Sensor

The HC-SR04 Ultrasonic Sensor consists of a multi-vibrator fixed at its base, which is a combination of a resonator and a vibrator. The ultrasonic waves generated by the vibration are delivered to the resonator. This sensor has two parts: the emitter, which produces a 40 kHz sound wave, and the detector, which detects the 40 kHz sound wave and sends an electrical signal back to the microcontroller. Ultrasonic sensors are primarily used as leveling sensors [13]. In our robot, an ultrasonic sensor is utilized to detect the amount of waste in the waste bin. They're also utilized to identify obstacles and rubbish. It is a transducer because it converts sound waves into electrical signals and vice versa. They are used for distance measurements by sending out ultrasonic waves and measuring the time it takes for the echo to return. Calculates distance to objects based on the speed of sound. The HC-SR04 ultrasonic sensors consist of 4 pins: VCC, Trigger, Echo, and GND. In the Robotic waste collector, an ultrasonic sensor provides real-time distance data to the ESP32, which uses this information to avoid collisions by adjusting the robot's path. Two Ultrasonic sensors have been fixed in front of the robot in such a way that any object that lies between the lower ultrasonic and the top ultrasonic sensor is detected as trash and the ultrasonic sensor placed on top is used for obstacle detection.

(f) Metal proximity sensor

The Metal proximity sensor (inductive proximity sensor) can detect metal targets near the sensor without any physical contact. It is placed in the dustbin manipulator, and it detects metal waste, and it is a close-range obstacle avoidance. When metal is detected, the sensor sends a signal to the microcontroller (ESP32), indicating that a metallic item is present. Based on the signal from the proximity sensor, the robotic waste collector will activate the servo motor responsible for bin rotation to separate metal from non-metal waste and segregate accordingly. This is used to detect if the trash picked is metal or non-metal.

(g) 12V Lead Acid Battery

The 12V Lead Acid Battery is the source of energy supply to the robotic waste collector. This battery[26] ensures that the system is adequately powered and runs optimally for

longer periods, by providing the continuous electrical power needed to effectively manage the robot and all its features. It is rechargeable and hence is affordable and enables sustainability.

(h) 9V General Purpose Battery

The 9V General Purpose Battery is utilized to supply power to smaller electronic components like the different types of sensors that require low-voltage conditions. It is a compact and versatile power supply unit for the ESP32 microcontroller and various sensors ensuring constant and more efficient power supply.

(i) IC7812 Voltage Regulator

The IC7812 Voltage Regulator is used to convert higher input voltages to a stable 12V output. It regulates the input voltage, for instance, a voltage higher than 12V, and provides an output of a steady voltage of 12V. It ensures that BO motors receive 12V voltages and prevents damage due to voltage fluctuation. It has Built-in thermal protection to guard against overheating by dissipating excess heat efficiently.

(j) IC7805 Voltage Regulator

The IC7805 Voltage Regulator provides a stable 5V output when the input voltage is more than 5V. It ensures consistent voltage is supplied to the components which require 5V for performance such as the ESP32 and other sensors. Its features include an output voltage of 5V, an input voltage range of 7-25V, current output up to 1.5A, short circuit protection, and reliable and stable voltage regulation.

3.1.3 Software requirements

(a) Arduino IDE

Arduino Integrated Development Environment (IDE) is used to write computer code in C or C++ and upload this code to the physical board. This software is used to program the robotic waste collector to autonomously identify, collect, segregate, and navigate the environment.

The Arduino IDE [16][17] features a simple text editor for writing code. It supports various libraries and provides syntax highlighting to aid in the development process. The IDE compiles the written code into a format that the microcontroller can understand. In this process, code is checked for any errors and converts the high-level code into machine language. The compiled code is uploaded to the microcontroller via a USB connection or over-the-air (OTA) for supported boards like the ESP32. This step transfers the instructions to the robot, enabling it to perform its tasks.

(b) Blynk Application

Blynk is a cloud service that enables users to control [14] and/or monitor their Arduino, Raspberry Pi, ESP32, and other microcontroller-based projects through an internet interface. It has an application that Android and iOS devices can use which allows controlling the hardware from a mobile device, additionally, it's flexible with widgets that can be added to the mobile application.

In this project, Blynk provides the ability to monitor and control robotic waste collectors through its mobile application. It provides remote control which enables the user to control some movements of the robot such as forward movement, backward movement, left or right. It helps in real-time monitoring of data that is from the ultrasonic sensors[25] indicating the distance measurement which is very useful for determining the presence of an obstacle and the bin level threshold. Through the application, users can switch the functioning mode of the robot on or off. They could also change the mode to manual or autonomous. It provides a friendly user experience.

It consists of three main elements that facilitate the interaction between the user and the robotic waste collector such as Blynk app builder, Blynk server and Blynk Libraries. Blynk App Builder tool helps users build interfaces using many widgets. These can include particularly buttons, sliders, and displays, as they are a convenient way to send commands to the hardware and receive responses. The Blynk Server can also be described as a middle link between the hardware and the mobile device, which controls the connection. It also serves the purpose of interconnection by facilitating the stream of data as well as the commands between the two. Blynk libraries are incorporated in the code to facilitate the connection with the Blynk Server, and they deal with establishing a connection to the server, transmitting data, and receiving commands.

This robotic waste collector uses a few libraries provided by Blynk which helps users to easily connect their mobile application with the hardware. The few libraries used in this project are BlynkSimpleEsp32.h which is useful for connecting ESP32 microcontroller to a cloud service named Blynk, it the middle link between the hardware and the Blynk application, WiFi.h and WiFiClient.h, libraries help the ESP32 to connect to the Wi-Fi network and for getting connect with Blynk's server through the internet and Servo.h library is used to control the dustbin manipulator and bin rotation.

3.2 Working Modes

3.2.1 Automatic Mode

In the automatic mode of our robotic waste collector, we implement obstacle detection and avoidance through a dual ultrasonic sensor setup. Two ultrasonic sensors, labeled A and B, are strategically set in two parallel rows at the front part of the robot. As our robot traverses its path, the ultrasonic sensors continuously scan the environment, aiming to discern between obstacles or objects and potential trash.

Ultrasonic sensor A is positioned in the lower part of the chassis, while ultrasonic sensor B is placed a few centimeters above sensor A as shown in Fig. 3.4. When both ultrasonic sensors [15] [19] detect an object or obstacle, the robot's response varies based on the nature of the obstruction. If the sensors (A and B) collectively identify a solid object or a wall, the robot initiates a turning mechanism to navigate around the obstacle, giving preference to sensor B. It then turns right and checks for obstacles. If an obstacle is detected, the robot returns to the center point and takes a left turn to explore further for trash. Likewise, if the obstacle is detected solely by ultrasonic B, the robot promptly turns and takes another path, executing the right-turn-and-check-center-left-turn process.

This design choice allows the system to give precedence to readings from sensor B, considering them more useful in identifying potential obstacles. Additionally, if an object is detected only by sensor A, it is considered to be trash, with the exception that in outdoor environments such as parks and gardens, where lengthy grasses are prevalent, the system may inaccurately identify them as trash.

This approach ensures that the robotic waste collector effectively navigates through its environment, distinguishing between obstacles that warrant avoidance and trash that merits collection.

Fig. 3.3 depicts the flow of the robot's movement in autonomous mode. Once the robot starts moving in the forward direction and the ultrasonic sensor closer to the chassis detects an object, then it is considered trash, and the trash is collected into the bin by the manipulator. If both the ultrasonic sensors (the one placed bottom closer to the chassis and the other one in the top) detect the object, then it is considered as obstacle. This is because we assume that the height of the trash to be less than the height of the robot's chassis. In this case after detecting the object as obstacle, the robot turns towards right direction and moves forward.

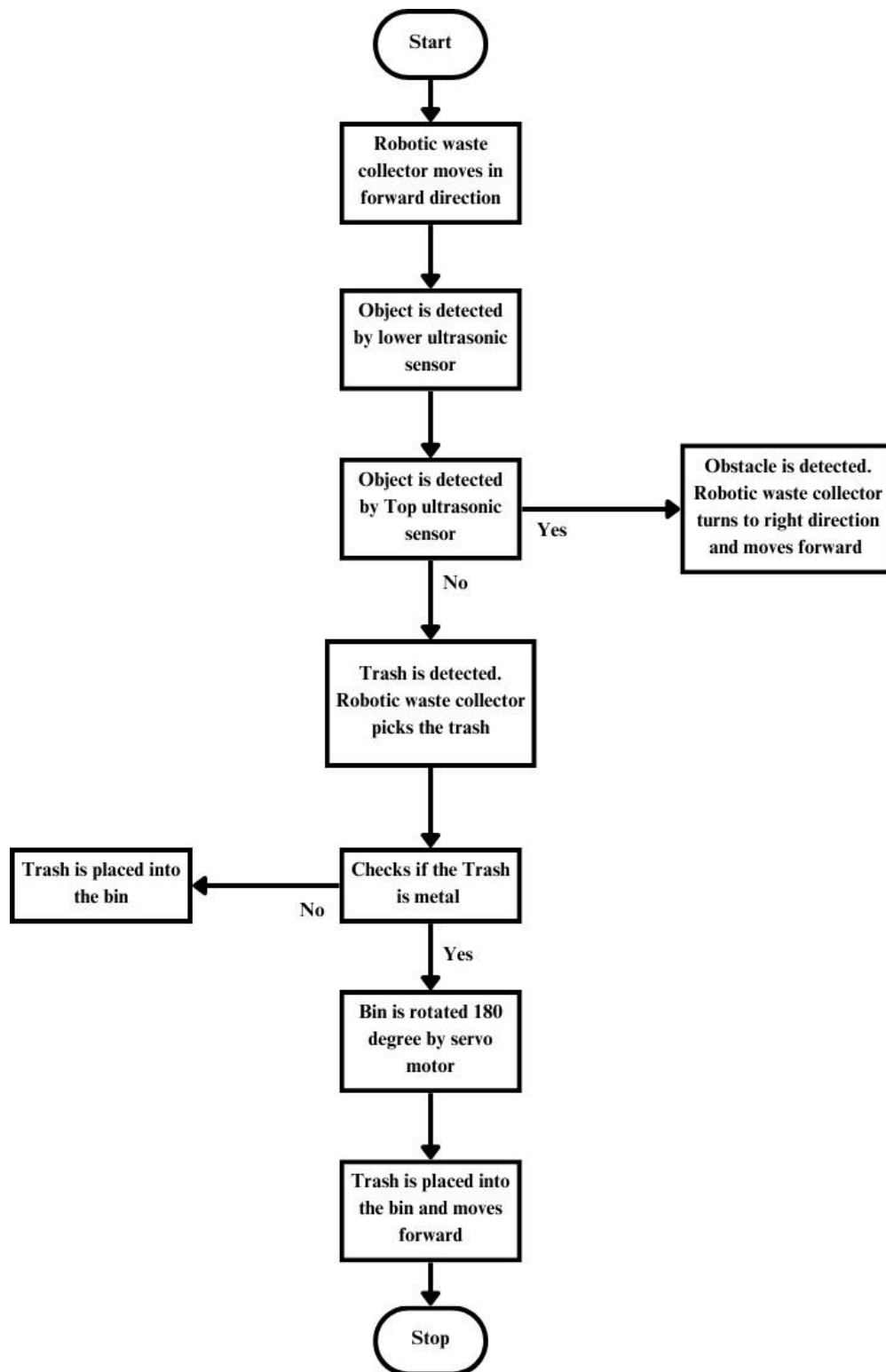


Fig. 3.3 Flowchart for Autonomous Movement of Robot

3.2.2 Manual Mode

In the manual mode, instead of relying on autonomous decision-making, the robot follows user controls transmitted through the Blynk app interface. Obstacle detection and

avoidance are integrated through the use of a mobile app using user commands. For real-time communication between the app and the robot, the ESP32 maintains a continuous connection with the Blynk server. The robotic waste collector utilizes a Wi-Fi module for communication which offers advantages over Bluetooth due to its longer range and enhanced connectivity stability. Wi-Fi operates on a higher frequency, enabling faster transmission of sensor data and user commands, crucial for real-time responsiveness.

The app empowers users to command the robot's movement [27]—forward, backward, left, and right—as well as control a servo-operated arm for efficient trash collection. While the ultrasonic sensors are still active, their data does control the robot's autonomous actions in manual mode [31]. Instead, the user's commands take precedence, enabling an interactive and user-driven experience. The user interface on the Blynk app is designed to include buttons, sliders, or any other widgets that will help the user control the robot. User inputs received from the Blynk app are mapped to specific actions to be performed by the robotic waste collector. For example, if the user presses a button to move the robot forward, corresponding with the use of code this command is interpreted by the ESP32 and controls the motors for forward movement accordingly.

3.3 Steps Involved

3.3.1 Trash Collection and Segregation

Before delving into the trash identification and collection process, it's crucial to note that whether the trash is collected autonomously or manually, the subsequent step involves determining if the trash is of metal or non-metal composition. The bot collects the waste trash from the ground using its modified dustbin manipulator (i.e., a dustbin cut into two equal halves). Each end of the cut dustbin is attached to the corresponding end of the manipulator.

(a) Trash Identification

Once the robot utilizes its arms to pick up the trash, the following step involves trash identification as shown in Fig. 3.4. Our project incorporates a metal proximity sensor within the robot's arm, i.e., inside the dustbin manipulator. The metal proximity sensor assists in identifying whether the trash is metallic or non-metallic.

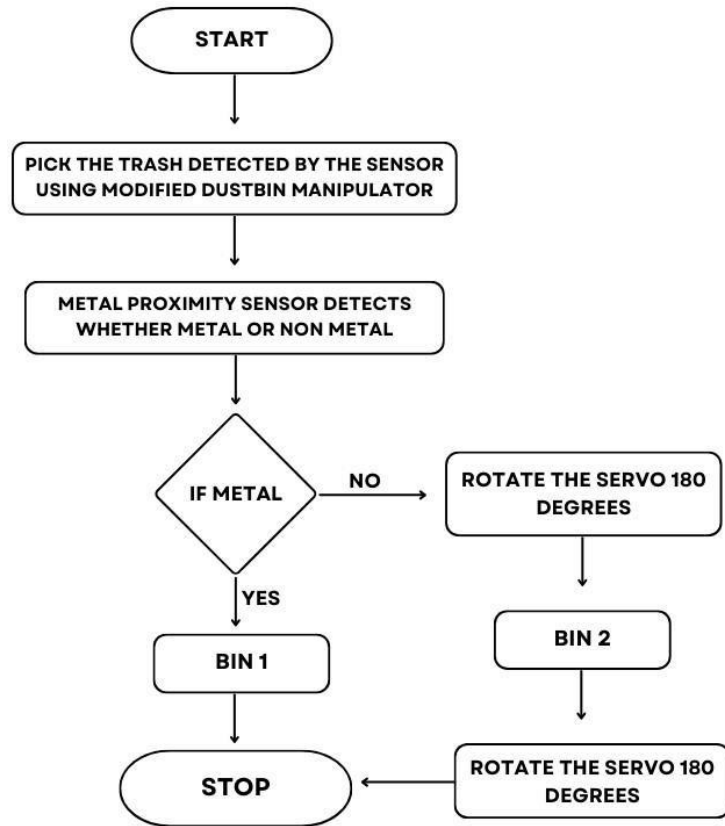


Fig. 3.4 Flowchart for Trash Identification

(b) Trash Collection

A bin is separated into two halves equally for non-metal and metal items, attached to the robot. After the trash is identified using the metal proximity sensor, if the waste is metallic, it is placed inside the bin. However, if the waste is non-metallic, the bin is rotated 180° to collect the non-metallic waste. Table 3-1 shows the commonly available trash found in public places that are detected by metal proximity as metal and non-metal. This process ensures that each type of waste is appropriately deposited in the corresponding section within the bin, promoting an organized and efficient waste collection system.

Table 3-1 Materials Detected by Inductive Sensor

Material	Inductive	Result
Plastic Cover	No change	Non-Metallic waste
Paper	No change	Non-Metallic waste
Metal Ring	Detects	Metallic waste
Banana Peel	No change	Non-Metallic waste
Glass Bottle	No change	Non-Metallic waste
Beverage Cans	Detects	Metallic waste
Aluminum foil	Detects	Metallic waste

3.3.2 Bin Level Monitoring Using Blynk App

The Bin Control[30] and Level Monitoring aspect of our waste management system is integral to ensuring efficient and timely waste disposal. The bin is divided into two sections, segregating collected scraps based on their material composition:

Section 1: For metal trash (one-half of the bin)

Section 2: For non-metal trash (the other half of the bin)

Continuous monitoring of the filled levels within both sections is achieved through the implementation of ultrasonic sensors. To ensure precise and accurate level monitoring, sensors are placed on the lid of the bin, each one on both segregated sections. These sensors accurately gauge the real-time levels of metal and non-metal trash present in the bin. Predetermined thresholds are set for each section of the bin. These thresholds represent the optimal fill capacity, ensuring the bin can handle waste effectively. The system considers the capacity constraints of the bin and ensures that it doesn't exceed its designated limits.

Once the predetermined threshold is crossed in either section, the system triggers notifications, promptly alerting the user to take necessary action. This timely notification serves as a proactive reminder to empty the bin. The implementation of this smart level monitoring system not only optimizes the usage of the bin but also offers users a hassle-free and efficient approach to waste management [18]. By providing real-time updates and proactive alerts [35], the system promotes a user-centric and environmentally conscious waste disposal process.

4. Result and Discussion

4.1 Simulation Results

Fig. 4.1 illustrates the simulation results of the DC motors (responsible for the robot's wheels movement) and servo motors utilized for various functions such as rotating the dustbin and handling the pick-up and carry of waste.

There are four BO motors shown in Fig. 4.1, each depicting the wheels placed on the chassis of the robot, enabling movement in different directions: forward ('F'), backward ('B'), left ('L'), right ('R'), and stop ('S'), as seen in Fig. 4.2.

Servo motors play a key role in controlling different parts of the robotic arm and the rotation of the dustbin. Servo motors A, in Fig. 4.1, pick up trash from the ground. They control the ends of the modified dustbin manipulator during the trash-picking process. Servo motors B, highlighted in Fig. 4.1, help carry waste from the ground to the bin. Servo motor C, shown in Fig. 4.1, rotates the dustbin for efficient metal and non-metal segregation. It's marked as 'W' (for waste placement) in the virtual terminal, as seen in Fig. 4.3.

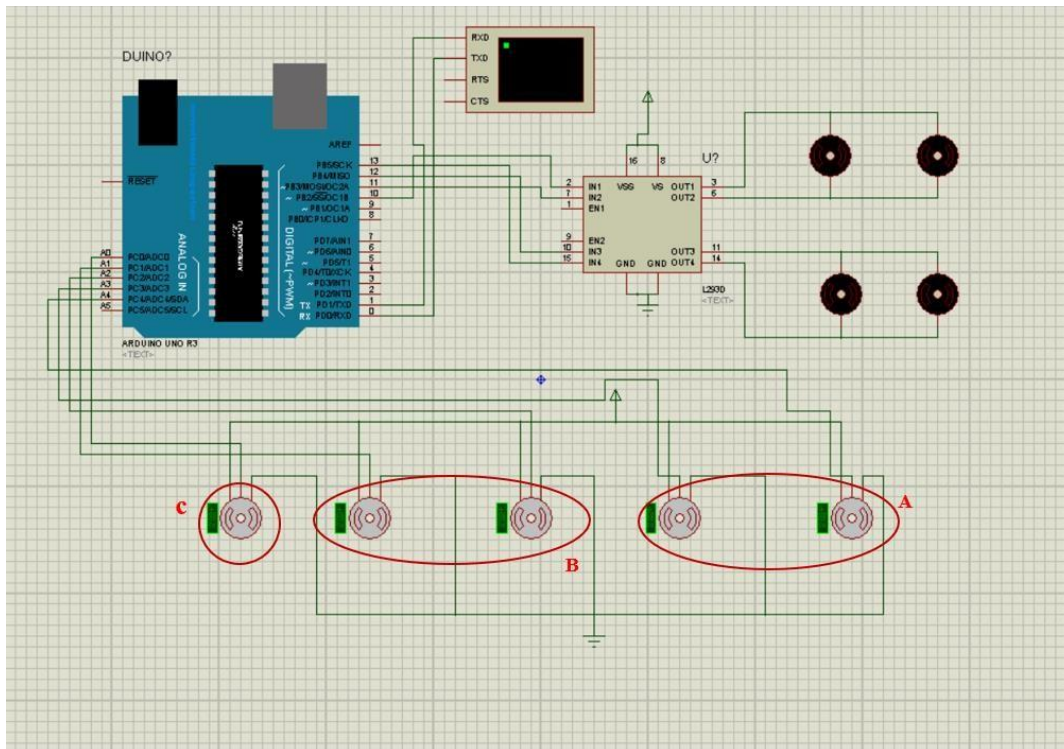


Fig. 4.1 Simulation of BO and Servo Motors

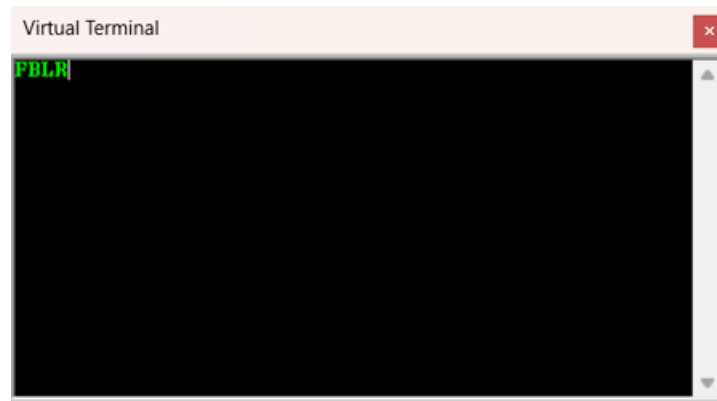


Fig. 4.2 Virtual Terminal for Wheel Rotation



Fig. 4.3 Virtual Terminal for Pick, Carry, Rotate

4.2 Sensor Calibration

The robotic waste management system's actuators and sensors were calibrated to guarantee accurate and dependable operation as shown in Table 4-1.

Table 4-1 Calibrated Sensor Values

S.NO	Component	Model	Specification	Threshold range	Usage
1	Servo motor 1	Tower pro SG-90 micro servo	0-180 degree	0-180	Rotation of dustbin
2	Servo motor 2	MG-996R	0-180 degree	0-90	Rotation to carry the wastage
3	Servo motor 3	MG-996R	0-180 degree	90-0	Rotation to Carry the wastage
4	Servo motor 4	MG-996R	0-180 degree	0-180	Arm rotation

5	Servo motor 5	MG-996R	0-180 degree	0-180	Arm rotation
6	Ultrasonic Sonic 1	HC-SR04	0-200cm distance	<50	Obstacle detection
7	Ultrasonic Sonic 1	HC-SR04	0-200cm distance	<30	Waste detection
8	Ultrasonic Sonic 1	HC-SR04	0-200cm distance	<20	Check metal Wastage level
9	Ultrasonic Sonic 1	HC-SR04	0-200cm distance	<20	Check Non-metal Wastage level

Ultrasonic Sensor Calibration: The ultrasonic sensors were calibrated to measure distances for obstacle detection and bin level monitoring precisely. This involved calculating the speed of sound in the current environmental conditions. The formula used for distance measurement was:

$$d = \frac{s \times t}{2} \quad (4.1)$$

Where,

d = Distance to an object

s = Speed of sound

t = Time taken by the ultrasonic signal to travel to the object and back

Once calibrated, the sensors could detect objects within a specified threshold limit. The ultrasonic sensor's Tx emitted a signal, which, upon striking an object, reflected to the Rx. The gap between the sensor and the object was determined as the distance, and when it reached the predefined threshold, the system initiated further actions.

Precise and controlled movements of the robotic arm and bin rotation were achieved by calibrating servo motors. The process of calibration included figuring out which rotational angle corresponded to specific control signals. Angle calibration does map control signals to specific angles of rotation. Verified the accuracy of the servo motors in achieving the desired angular positions.

4.3 Robot Chassis Setup

The robot's chassis setup is shown in Fig.4.4 and Fig.4.5. The robot's chassis is made of plywood which is around 25.3 cm long and has four wheels at the bottom. BO motors are attached to each of the four wheels. Two wooden boards are designed with a center gap to securely hold the ESP32, Motor Driver, and connecting wires. A hole in the middle of the

lower board is created to allow the wires to pass through neatly to power the BO Motors. On the top board, there's a servo motor that is placed with a circular wooden platform on which a bin can be placed. This setup allows the servo motor to rotate the bin, facilitating the separation of metal and non-metal items. There are two ultrasonic sensors present in front of the chassis to facilitate obstacle detection and avoidance.

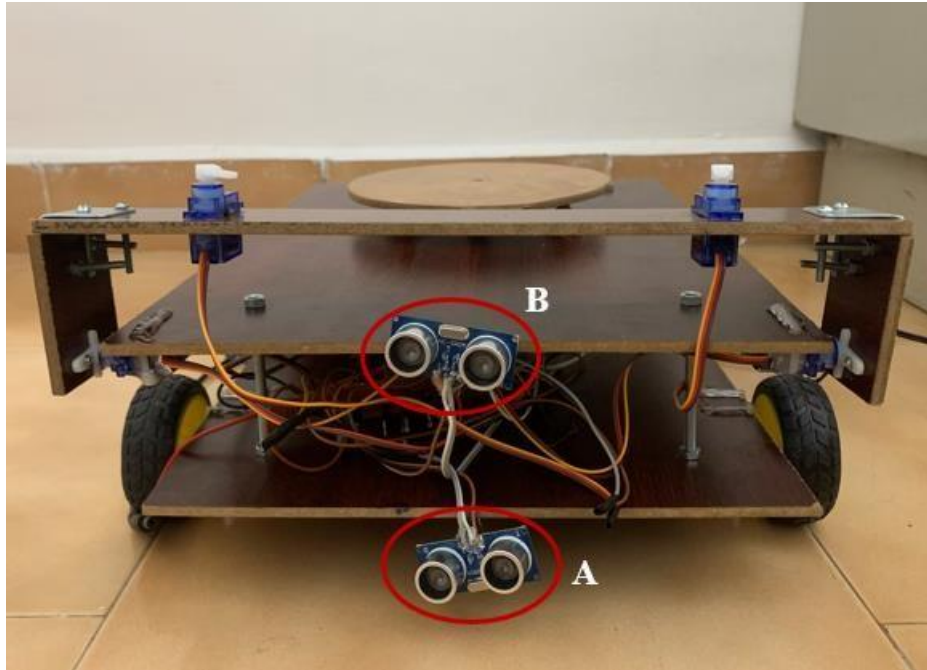


Fig. 4.4 Robot Chassis setup – Front view



Fig. 4.5 Robot Chassis setup – Side view

4.4 Blynk App Interface

In Fig. 4.6 of the Blynk app interface, the "obstacle" widget is highlighted, representing ultrasonic sensor B from Fig. 4.4 for obstacle detection. The "wastage"

widget corresponds to ultrasonic sensor A, which detects trash. This widget shows that ultrasonic sensor located at the bottom of the robot has found a piece of waste. It is important to distinguish between waste and obstacles and for this purpose the sensor determines their positions in relation to each other. When the bottom sensor has detected an object, but the top has not, this is considered waste. When both the sensors pick something, they are considered as an obstacle.

Metal Waste Level can be seen using Bottom Left Widget. This widget regulates the levels of metal waste being dumped in the dustbin. The gauge gives an operation indication of the status of metal wastes so that the operator is informed that the bin is full and requires it to be disposed. Non-Metal Waste Level can be monitored using Bottom Right Widget. Similarly, for metal waste level gauge, this gauge shows the non-metal waste that is contained in the dustbin. This information is very essential in as far as segregation and management of wastes is concerned as well as its disposal.

Waste can be collected using “pick”, “carry”, “Waste bin” Button. Pick button when triggered the robot’s manipulator take the waste from the ground. The carry button instructs the robot to carry the trash from the ground to the bin. In this process, the trash is segregated using the metal proximity sensor placed in dustbin manipulator. The waste button controls robots to place the trash into the respective bin based on the trash detected by metal proximity sensor.

“Automatic Mode On/Off” button switches the robot between automatic and manual waste collection mode. When the button is in the “On” position the robot assumes full control and makes decisions on when and where to move and collect trash on its own using its built-in sensors. With the power switch in the “Off” position control of the robot is by the directional and task buttons.

“Forward”, “Backward”, “Left”, “right” buttons are used as directional control widgets. These four buttons are used for determining the motion of the robot in the mentioned directions. Depending on the position of the button pressed, the robot translates in the desired directions (forward, backward, towards the left, or towards the right).

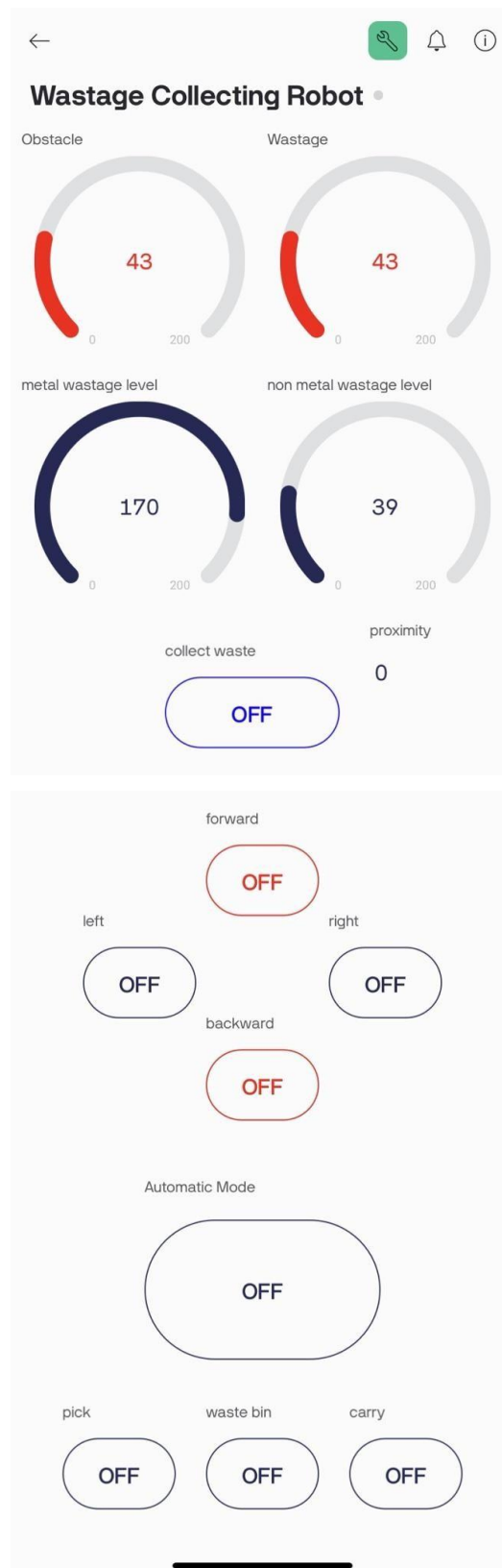


Fig. 4.6 Blynk Application Interface

4.5 Integration of Manipulator with Robot Chassis



Fig. 4.7 Resting Position of Robotic Waste collector



Fig. 4.8 Collecting of waste using Manipulator

The one shown in Fig. 4.7 is the resting position of the robot. This robot on activation initiates the movement of picking trash and is expected to dispose it in a bin with a label either “METAL” or “NON-METAL”, when the work is done, the robot returns to this state in readiness to perform the next cycle of work. The one shown in Fig. 4.8 is the position of the robot while collecting the waste. The robot's mechanism is actively scooping up the trash and preparing to deposit it into the bin. This demonstrates the robot's functional capability in waste collection before it returns to its resting position.

4.6 Trails Conducted for Waste Collection and Segregation

Table 4-2 Trails for manual waste collection

Material	No. of times Collected	No. of times not Collected
Paper ball	8	2
Aluminium Foil	9	1

The frequency of collection for the two materials is shown in the Table 4-2. Eight paper balls were successfully collected, and two were not. In the instance of aluminium foil, nine collections were made successfully, and one was unsuccessful.

Table 4-3 Trails for dropping waste into the bin

Material	No. of times dropped into the bin	No. of times not dropped into the bin
Paper ball	6	2
Aluminium Foil	7	2

The success and failure percentages of dumping two distinct materials into a bin are shown in the Tale 4-3. Six times, paper balls were dropped into the bin successfully; the othertwo times, they were not. In the instance of aluminium foil, it was dropped into the bin seven times successfully and twice unsuccessfully.

Table 4-4 Trails for segregation in manual mode

Material	No. of times correctly segregated	No. of times not segregated correctly
Paper ball	6	0
Aluminium Foil	4	3

The accuracy of segregation for two distinct materials is shown in the Table 4-4. Paper balls havea perfect segregation record of six times and a zero-error rate. There have

been four instances of flawless segregation and three instances of faulty segregation in the case of aluminium foil.

Table 4-5 Trails for automatic waste collection

Material	No. of times Collected	No. of times not Collected
Paper ball	7	3
Aluminium Foil	7	3
Coke Tin	8	2
Small Plastic Bottles	7	3

The collecting efficiency for each of the four material categories is shown in the Table 4-5. Coke tin was collected successfully eight times and not collected twice, paper balls were collected seven times and not collected three times, aluminium foil was collected seven times and not collected three times, and small plastic bottles were collected seven times and not collected three times.

Table 4-6 Trails for dropping waste into specific bin

Material	No. of times dropped into bin	No. of times not dropped into the bin
Paper ball	6	1
Aluminium Foil	7	0
Coke Tin	8	0
Small Plastic Bottles	5	2

The success and failure rates of putting four distinct materials in a bin are shown in the Table 4-6. Paper balls were disposed of correctly six times and erroneously one time, and small plastic bottles were disposed of correctly five times and incorrectly two times. Aluminium foil and Coke tins have flawless disposal records with a zero-failure rate.

Table 4-7 Trails for segregation in automatic mode

Material	No. of times correctly segregated	No. of times not segregated correctly
Paper ball	6	0
Aluminium Foil	6	1
Coke Tin	7	1
Small Plastic Bottles	4	1

The Table 4-7 displays the precision of segregation for four different materials. Paper balls have a zero-mistake rate and a perfect segregation record of six times. Coke tin was separated seven times without being separated once, aluminum foil was separated six times and not once, and little plastic bottles were separated four times without being separated once.

4.7 Challenges Encountered

Developing the IoT-enhanced robotic waste collection faced a number of technical and operational obstacles. These issues underlined the current design's limitations and the complexities of developing an effective, dependable, and diverse waste management system. Here, we'll go over the main issues encountered during the development and testing phases.

While the robot was preferable to drive on tiles and plain surfaces [32], it was expected that the performance would decrease on other types of ground. It was maneuverable and stable especially when moving on tiles and flat surfaces making distinct turns and stops. However, these controlled environments are not capable of exposing the numerous real-life scenarios when garbage collection is needed. Grass and uneven terrain has constrained the movement of the robot only on grass that was less than half the diameter of the wheels of the robot. One of the drawbacks of this constraint was its effectiveness in areas of long or untidy grass or terrain which in most of the outdoor proved to be very limited. Solving this problem would require the enhancement of the robot's wheel structure and the firmness of the ground contact.

The dustbin attached to robot lacked the capacity to take a large amount of trash, it could only accommodate 150-200 grams of the rubbish. Due to the dustbin loading capacity of the robot utilized being limited, it called for constant emptying of the dustbin which interfered with operation and made the system less efficient. This task justifies the need for a bigger or more effective waste container that can enhance the robot's operational self-sufficiency.

The accumulation of the metallic as well as the non-metal items obstructed the segregation of trash. If the robot were to pick both the metal and the non-metal things at the same time, it may put them together in the bin depending on what the proximity sensor first detected.

This led to what can be described as wrong categorization. Based on the evaluation results and analysis, it is necessary to upgrade the sensing algorithm and the mechanism of segregation.

The current segregation method of trash was only able to segregate waste following a metallic attribute. Two of the important categories of wastes had not been sorted in an efficient manner by the robot, namely dry and wet wastes. This requires a better trash identification framework that is better suited to trash types that are not categorized as Metallics and Non-Metallics. And could be further classified as dry and wet wastes, which allows us to be used in various fields/ industries.

New challenges posed by the change of environment from static laboratory conditions to the natural environment were a challenge to the robot. Sensors and other external environmental conditions such as rain, extreme temperature, and wind may pose some effect on the robot's accuracy of movement and waste classification, by causing either a failure in operations or lowered effectiveness.

5. Conclusion and Future Scope

In conclusion, the provided robotic waste management system powered by the ESP32 microcontroller provides a novel and efficient solution to the issues of metal waste collection and on-site segregation. The system focuses on improved material sorting, specifically separating metallic and non-metallic materials. Key components include metal proximity sensor for trash classification and the incorporation of the Blynk Android application for user experience.

In automated mode, the robot navigates through its environment, identifying obstacles and collecting the trash. The combination of ultrasonic and metal proximity sensors provides accurate waste detection. The manual mode allows users to manage the robot's movement and arm for trash collection through the Blynk app. By offering a simple-to-use platform for real-time interaction with the robotic waste collector, the Blynk app interface improves user experience [36].

Future scope for this project could include implementing machine learning and deep learning to segregate different waste and categorizing the waste into more categories than just metal and non-metal and smart obstacle detection. Path planning algorithm could be implemented for efficiently navigating through unknown terrain rather than solely depending on ultrasonic sensors. Another aspect could be to reduce the cost of building this robot by effectively selecting and minimizing the use of different sensors and materials.

Overall, Robotic Trash collector contributes to environmentally friendly waste management practices by addressing the challenges associated with metal waste. The incorporation of IoT technology, automation, and user-friendly interfaces provides foundations for more long-term and sustainable recycling activities, promoting a cleaner and healthier environment. The systematic approach presented in this work paves the way for smarter and more efficient waste management solutions in the context of growing environmental concerns.

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